

## Introduction

We begin by finishing off our conversation with the Euclidean algorithm by talking about how to solve Linear Diophantine Equations. Afterwards we will turn our focus to prime numbers and their properties. It turns out we can even redefine what it means to be prime, in terms of divisibility.

### Learning Objectives

- Define a linear Diophantine equation and find all the solutions to such an equation.
- Prove basic facts about prime numbers.
- State the divisibility definition of prime numbers and prove its equivalence to the distinct factors definition.
- State and use the Fundamental Theorem of Arithmetic.

## Linear Diophantine Equations

### Proposition 1

If  $a \mid bc$  and  $\gcd(a, b) = 1$  then  $a \mid c$ .

**Exercise 2** Prove Proposition 1.

### Definition 3

Given  $a, b, d \in \mathbb{Z}$ , a Linear Diophantine Equation (in two variables) is any equation of the form

$$ax + by = d.$$

Certainly there are rational solutions to this equation, but we are more interested in finding integer solutions.

### Theorem 4

If  $a, b, c \in \mathbb{Z}$ , then  $ax + by = c$  has a solution if and only if  $\gcd(a, b) \mid c$ .

**Exercise 5** Prove Theorem 4.

### Example 6

Find a solution to the Diophantine equation  $504x + 1155y = 42$ .

*Solution.* By applying the Euclidean algorithm, we get

$$1155 = 504(2) + 147$$

$$504 = 147(3) + 63$$

$$147 = 63(2) + 21$$

$$63 = 21(3) + 0.$$

Thus  $\gcd(504, 1155) = 21$ , and since  $21 \mid 42$  we know from Theorem 4 that our equation has a solution.

As when finding solutions to Bézout's Identity, we will write the last line as  $21 = 147 + 63(-2)$ . We now solve each remaining equation for the remainder to find  $63 = 504 + 147(-3)$  and  $147 = 1155 + 504(-2)$ . Substituting these we get

$$\begin{aligned} 21 &= 147 + 63(-2) \\ &= 147 + [504 + 147(-3)](-2) && \text{from } 63 = 504 + 147(-3) \\ &= 147(7) + 504(-2) \\ &= [1155 + 504(-2)](7) + 504(-2) && \text{from } 147 = 1155 + 504(-2) \\ &= 1155(7) + 504(-16). \end{aligned}$$

This is not the desired solution though, so we multiply through by 2 to get

$$504(-32) + 1155(14) = 42$$

as required. ■

**Exercise 7** Find a solution to the linear Diophantine equation  $1053x + 481y = 39$ .

There are in fact infinitely many solutions to a solvable Diophantine equation. How do we find them in general?

**Proposition 8**

Suppose that  $a, b, c \in \mathbb{Z}$  and  $(x_0, y_0)$  satisfy  $ax_0 + by_0 = c$ . If  $d = \gcd(a, b) \neq 0$  then the general solution to the Diophantine equation  $ax + by = c$  is given by

$$x = x_0 + n\frac{b}{d}, \quad y = y_0 - n\frac{a}{d}, \quad \text{for all } n \in \mathbb{Z}.$$

Note that it does not matter whether you put the minus sign in the  $x$  term or the  $y$  term, so long as they are opposing signs.

*Proof.* Subtract the equations  $ax_0 + by_0 = c$  and  $ax + by = c$  to find that  $a(x - x_0) + b(y - y_0) = 0$ . Divide through by  $d$  and re-arrange to find that

$$\frac{a}{d}(x - x_0) = -\frac{b}{d}(y - y_0).$$

We showed in a previous exercise that  $\gcd(a/d, b/d) = 1$ . Moreover,  $a/d$  divides the left hand side, and so must also divide the right hand side. Since  $\gcd(a/d, b/d) = 1$ , then by Proposition 1 we know that  $(a/d) \mid (y - y_0)$ . Similarly,  $(b/d) \mid (x - x_0)$ , so there exists  $n \in \mathbb{Z}$  such that  $x - x_0 = n(b/d)$  so that

$$-\frac{b}{d}(y - y_0) = \frac{a}{d}(x - x_0) = n\frac{a}{d}\frac{b}{d} \implies bd(y_0 - y) = anb.$$

In both cases, we can solve for  $x$  and  $y$  to find that

$$x = x_0 + n\frac{b}{d}, \quad y = y_0 - n\frac{a}{d}.$$

Now in fact every such  $n$  works, since

$$ax + by = a\left(x_0 + n\frac{b}{d}\right) + b\left(y_0 - n\frac{a}{d}\right) = (ax_0 + by_0) + \left(\frac{anb}{d} - \frac{anb}{d}\right) = ax_0 + by_0 = c. \quad \square$$

### Example 9

Find the general solutions to  $504x + 1155y = 42$ .

*Solution.* We have already found a particular solution; namely,  $504(-32) + 1155(14) = 42$ . Furthermore,  $d = \gcd(504, 1155) = 21$  and so the general solutions are of the form

$$x = -32 + n\frac{1155}{21} = -32 + 55n, \quad y = 14 - n\frac{504}{21} = 14 - 24n. \quad \blacksquare$$

It is sometimes easier to normalize by the greatest common divisor before starting. Take our carry-through example, where we want to solve  $504x + 1155y = 42$ . Dividing through by the greatest common divisor  $d = \gcd(504, 1155) = 21$  we get the equation

$$24x + 55y = 2.$$

Since  $\gcd(24, 55) = 1$  we can find a solutions to  $24x + 55y = 1$ , then multiply by 2. To do this, we again use the Euclidean algorithm to find

$$55 = 24(2) + 7$$

$$24 = 7(3) + 3$$

$$7 = 3(2) + 1$$

$$3 = 3(1) + 0$$

(where of course we already knew the greatest common divisor is 1). Now working backwards

$$\begin{aligned} 1 &= 7 + 3(-2) \\ &= 7 + [24 + 7(-3)](-2) \\ &= 7(7) + 24(-2) \\ &= [55 + 24(-2)](7) + 24(-2) \\ &= 55(7) + 24(-16). \end{aligned}$$

Multiplying by 2 gives us our solution  $24(-32) + 55(14) = 2$ . Notice that  $(-32, 14)$  are the same as the solutions we found above, as should be the case. Moreover, the general solutions are easily read off as  $x = -32 + 55n$  and  $y = 14 - 24n$ , which is again the same solution we found above.

**Exercise 10** Find all solutions to the linear Diophantine equation  $1053x + 481y = 39$ .

**Example 11**

Find all non-negative solutions to  $504x + 1155y = 42$ , if they exist.

*Solution.* We know that the general solutions are of the form  $x = -32 + 55n$  and  $y = 14 - 24n$ . We require both numbers to be positive, giving  $-32 + 55n > 0$  and  $14 - 24n > 0$ , which we can solve for  $n$  to get

$$\frac{32}{55} < n < \frac{14}{24}.$$

There are no integers which satisfy this equation, so there are no non-negative solution. ■

## Primes

You're likely already familiar with the classical definition of a prime number, but we'll include it here again just for clarity.

**Definition 12**

A *prime* number is any positive integer greater than 1 that cannot be written as the product of two smaller natural numbers.

**Theorem 13**

A number  $p \in \mathbb{Z}^+$  is prime if and only if for any  $a, b \in \mathbb{Z}$ , if  $p \mid ab$  then  $p \mid a$  or  $p \mid b$ .

**Exercise 14** Write down Theorem 13 using propositional logic.

*Proof.* ( $\Rightarrow$ ) Assume that  $p$  is prime and that  $p \mid ab$ . If  $p \mid a$  we are done, so assume that  $p$  does not divide  $a$ , in which case  $\gcd(a, p) = 1$ . By Theorem 1, it then follows that  $p \mid b$ .

( $\Leftarrow$ ) Conversely, we will proceed by the contrapositive; that is, we will show that if  $p$  is not prime, then there exists  $a, b$  such that  $p \mid ab$  but neither  $p \mid a$  nor  $p \mid b$ . Assume that  $p$  is not a prime, so it is necessarily composite. We can thus write  $p = rs$  for  $1 < r \leq s < p$ . Hence  $p \mid rs$  but  $p$  divides neither  $r$  nor  $s$ , □

This fact is equivalent to being a prime number, and is in fact the definition of prime in higher mathematics. While we do not yet have the tools to prove it, you can probably believe that if  $p$  is a prime and  $p \mid a_1 a_2 \cdots a_n$  then  $p \mid a_i$  for some  $i \in \{1, \dots, n\}$ .

**Exercise 15** Show that primality is necessary for Theorem 13; that is, give an example of numbers  $p, a, b$  such that  $p$  is not prime,  $p \mid ab$ , but  $p \nmid a$  and  $p \nmid b$ .

**Exercise 16** Show that if  $n \in \mathbb{Z}$  and  $p$  is prime, then  $p \mid n$  if and only if  $p \mid n^2$ .

**Exercise 17** Show that if  $p$  is a prime, then  $\sqrt{p}$  is irrational. *Hint:* Emulate the proof that  $\sqrt{2}$  is irrational. It is almost exactly the same proof.

### Theorem 18: The Fundamental Theorem of Arithmetic

Every positive integer greater than 1 can be uniquely expressed as a product of primes.

*Proof.* Let's first show that every number can be written as a product of primes, after which we'll show that the product is unique. For the sake of contradiction, suppose that the result is not true, and define

$$S = \left\{ n \in \mathbb{Z} : \begin{array}{l} n > 1 \text{ and } n \text{ cannot be written} \\ \text{as a product of primes} \end{array} \right\} \subseteq \mathbb{Z}^+.$$

This set is non-empty by assumption, and so by the Well-Ordering Principle,  $S$  has a least element, say  $k$ . Now  $k$  cannot be prime (why?) and so necessarily there exist numbers  $1 < r, s < k$  such that  $k = rs$ . Since  $k$  is the minimal element of  $S$ , this means that both  $r$  and  $s$  can be written as products of primes, which in turn shows that  $k$  can also be written as a product of primes. This is a contradiction, and so  $S$  must be empty. That is, every positive integer greater than 1 is a product of primes.

It remains to show that this decomposition is unique. For the sake of contradiction, assume that there exists some integer  $n > 1$  which can be expressed with two different prime factorizations, say

$$n = p_1 \cdots p_n = q_1 \cdots q_m,$$

for some choice of primes  $p_i$  and  $q_j$ . Since  $p_1 \mid n$ , it must also be the case that  $p_1 \mid q_1 \cdots q_m$ . Since  $p_1$  is a prime, by Theorem 13 we must have  $p_1 \mid q_j$  for some  $j$ . By reordering if necessary, let this be  $q_1$ . Since  $q_1$  is also prime, the only way  $p_1 \mid q_1$  is if  $p_1 = q_1$ . Cancelling  $p_1$  and  $q_1$  thus gives

$$p_2 p_3 \cdots p_n = q_2 q_3 \cdots q_m.$$

We repeat the same argument above, deducing that  $p_2 = q_2$ ,  $p_3 = q_3$ , and generally that  $p_i = q_i$ . Hence  $n = m$  and, up to reordering the factors, the prime decomposition is unique.  $\square$

### Example 19

The number  $\log_{36}(105)$  is irrational.

*Solution.* For the sake of contradiction, assume  $\log_{36}(105) = p/q$  where  $\gcd(p, q) = 1$ . By definition of the logarithm,  $36^{p/q} = 105$ , or equivalently  $36^p = 105^q$ . Factoring 36 and 105 into primes gives

$$(2^2 \cdot 3^2)^p = (3 \cdot 5 \cdot 7)^q \Rightarrow 2^{2p} 3^{2p} = 3^q 5^q 7^q.$$

By the Fundamental Theorem of Arithmetic, it must be the case that

$$2p = 0, \quad 2p = q, \quad \text{and} \quad q = 0,$$

for which the only solution is  $p = q = 0$ . However, the number  $0/0$  is not rational, so this is a contradiction.  $\blacksquare$

### Theorem 20: Euclid's Proof of the Infinitude of Primes

There are infinitely many prime numbers.

*Proof.* For the sake of contradiction, suppose there are only finitely many prime numbers and enumerate them all as  $P = \{p_1, p_2, \dots, p_n\}$ . Define  $x = p_1 p_2 \cdots p_n + 1$ . Since  $x$  is larger than any of the elements of  $P$ , it cannot be prime itself. We claim that  $x$  cannot be written as a product of primes, which would contradict the Fundamental Theorem of Arithmetic.

To see this, note that if  $x$  were written as a product of primes, then for some fixed prime  $p_k$  we must have  $p_k \mid x$ . But then  $p_k \mid (p_1 p_2 \cdots p_n + 1)$  and  $p_k \mid p_1 p_2 \cdots p_n$ , so it must be the case that  $p_k \mid 1$ . This is clearly not possible, so  $x$  cannot be written as a product of primes, which yields our contradiction. We conclude that there are infinitely many primes.  $\square$

## Exercises

1. For each Linear Diophantine Equation (LDE)  $ax + by = c$  below, determine
  - i. A single solution to the LDE.
  - ii. All solutions to the LDE.
  - iii. All non-negative solutions to the LDE, if they exist.
  - (a)  $221x + 143y = 13$
  - (b)  $286x + 121y = 33$
  - (c)  $1700x + 867y = 17$

(d)  $1232x + 553y = 63$

2. Determine how many 0's occur at the end of  $100!$ . (Recall that  $n! = n(n-1)(n-2)\cdots 3\cdot 2\cdot 1$ ).
3. (a) Let  $a \in \mathbb{N}$  and write  $a$  in terms of its prime factorization as  $a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$ , where  $p_1, p_2, \dots, p_n$  are distinct primes. Show that  $b \in \mathbb{N}$  divides  $a$  if and only if  $b$  has a prime factorization of the form  $b = p_1^{d_1} p_2^{d_2} \cdots p_n^{d_n}$ , where  $0 \leq d_i \leq a_i$  for  $i = 1, \dots, n$ .  
 (b) Suppose that  $a$  and  $b$  are natural numbers. Using a common set of distinct primes  $p_1, \dots, p_n$ , write  $a$  and  $b$  as

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n} \quad \text{and} \quad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n},$$

where some of the  $a_i$  and  $b_i$  may be zero as necessary. Show that

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}.$$

- (c) Determine the GCD of  $a = 9^4 \cdot 10^2 = 656100$  and  $b = 12^3 = 1728$ .
4. (a) By imitating Euclid's proof, show that there are infinitely many primes of the form  $4k + 3, k \in \mathbb{Z}^+$ .  
 (b) Once you have a valid proof for numbers of the form  $4k + 3, k \in \mathbb{Z}^+$ , show that your proof does not work for primes of the form  $4k + 1, k \in \mathbb{Z}^+$ .
5. (a) Determine the number of positive divisors of the number 360.  
 (b) In general, given a number  $n$  written in its prime factorization as  $n = p_1^{d_1} \cdots p_n^{d_n}$ , how many positive divisors are there for  $n$ ?  
 (c) If  $n$  is written in its prime factorization as  $n = p_1^{d_1} \cdots p_n^{d_n}$ , determine a condition on the  $d_i$  that ensures  $p$  is a perfect square; namely, that  $p = a^2$  for some number  $a$ .